

# Kshitish Pandit

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## Summary

Computer Science (AI) student with a track record of shipping real-time computer vision applications – from webcam-based safety monitoring to biometric and pose-tracking systems – using OpenCV, Dlib, and deep learning models.

## Featured Computer Vision Projects

### Real-Time Drowsiness Detection

*Python, Dlib, OpenCV*

- Built a driver-alertness system tracking 68 facial landmarks in real time via webcam to detect prolonged eye closure
- Triggered a non-blocking audio alert (via threading) when eyes remained closed for 2+ seconds, avoiding UI freezes

### Kitchen Hazard Detection System

*Multimodal AI — Python, OpenCV, Gemini API*

- Built a real-time kitchen safety monitor combining live video feeds with multimodal AI analysis
- Converted raw visual signals into clear, actionable hazard alerts for end users

### Exercise Tracker

*Python, OpenCV, Pose Estimation*

- Built rep-counting trackers for push-ups, squats, bicep curls, planks, and jumping jacks from video, sharing a common base-tracker architecture
- Output annotated video plus structured JSON rep/form data for each exercise session

### Face Recognition & Analysis System

*Python, TensorFlow, DeepFace, OpenCV*

- Built a real-time face recognition pipeline that identifies known individuals from a webcam feed
- Integrated DeepFace to analyze age, gender, emotion, and race attributes alongside recognition

### Image Captioning System

*Python, InceptionV3, LSTM with Attention*

- Combined CNN-based feature extraction (InceptionV3) with an attention-based language decoder to generate natural-language image captions on MS COCO

### Circle Catch Game

*Python, OpenCV, Haar Cascades*

- Built an interactive hand-detection game using Haar Cascade classifiers, with real-time scoring and a game timer

## Technical Skills

- **Languages:** Python
- **Computer Vision:** OpenCV, Dlib, Haar Cascades, DeepFace, MediaPipe
- **Deep Learning:** TensorFlow, Keras, CNNs, attention-based sequence models
- **Other:** NumPy, real-time video/webcam pipelines, multithreaded alerting

## Certifications

- Diploma in Data Science and Machine Learning – Vrit Technologies
- Natural Language Processing with HuggingFace Transformers

## **Education**

**Softwarica College of IT and E-Commerce**, Kathmandu  
Bachelor's in Computer Science (Artificial Intelligence focus)